SIMATS ENGINEERING

ASSESSMENT TOOL 2: Research Review & Analysis

COURSE NAME:	**ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS**	**COURSE CODE:**	**MLA0101**

COURSE OUTCOME COVERED
CO4: *Develop intelligent software agents using suitable agent architectures and learning techniques.*(BTL 3)
Assessment Tool Weightage: 30%

Total Marks: 100 **Time: 90 Mins**
No. of Questions: 1

Annexure A

Research Review & Analysis

Code of Conduct:

I K.Teja Sri(192525249) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

K. Teja Sri

Signature of the student:

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Domain Understanding	10%	
2. Literature Search and Source Selection	10%	
3. Literature Review and Synthesis	15%	
4. Comparative Analysis of AI Techniques	10%	
5. Critical Analysis of Research Findings	10%	
6. Research Gap Identification	10%	
7. Proposed Research Direction	10%	
8. Research Methodology / Analysis Framework	5%	
9. Experimental / Evidence-Based Validation	10%	
10. Result Analysis and Interpretation	5%	
11. Research Paper Quality and Referencing	10%	
12. Technical Presentation and Research Defense	5%	
Total	100%	

Signature of the Faculty

Total Marks Awarded

ANSWER ANY ONE QUESTION

S.No.	Research Review & Analysis Question	Primary Topic
1	**Decision Tree Learning:** Review recent research on ID3, C4.5, and CART. Compare their attribute-selection methods, pruning strategies, interpretability, computational complexity, and classification performance. Identify a research gap and suggest a possible improvement.	Decision Trees
2	**Neural Networks vs Statistical Learning:** Review and compare research on neural-network and statistical-learning approaches for prediction and classification. Analyze their performance, data requirements, generalization, interpretability, computational cost, and suitability for different applications.	Statistical Learning & Neural Networks
3	**Kernel Methods:** Analyze research on SVM and kernel methods for non-linear classification. Compare linear, polynomial, and RBF kernels and evaluate their advantages and limitations compared with neural-network-based approaches.	Kernel Methods
4	**Reinforcement Learning:** Review research on Q-Learning, SARSA, and Deep Q-Networks. Compare their learning mechanisms, exploration strategies, convergence, computational requirements, and real-world applications. Identify challenges and research gaps.	Reinforcement Learning
5	**Explainable and Generalizable AI:** Review research on Inductive Learning and Explanation-Based Learning and analyze how these approaches contribute to generalization and explainability in AI. Identify current limitations and propose future research directions.	Inductive & Explanation-Based Learning

Structure of the Report:

S.No.	Paper Section	Expected Content
1	**Title**	Specific and concise title related to the selected research question/topic.
2	**Abstract**	Brief summary of the research problem, review methodology, major findings, research gap, and conclusion.
3	**Keywords**	4–6 important keywords related to the selected topic.
4	**1. Introduction**	Introduce the AI topic, background, importance, motivation, and scope of the review.
5	**2. Research Problem and Objectives**	Clearly state the research problem and specific objectives of the review.
6	**3. Literature Review**	Review relevant research papers and summarize major approaches, methodologies, findings, and contributions.
7	**4. Comparative Analysis**	Compare the reviewed approaches based on suitable parameters such as accuracy, complexity, interpretability, computational cost, scalability, and generalization.
8	**5. Critical Analysis**	Discuss strengths, weaknesses, limitations, assumptions, and practical challenges identified across the reviewed studies.
9	**6. Research Gap**	Identify unresolved problems, limitations in existing research, and areas requiring further investigation.
10	**7. Proposed Research Direction**	Suggest possible solutions, improvements, or future research directions based on the identified gap.
11	**8. Discussion**	Discuss the overall insights obtained from the review and their implications for AI research and applications.
12	**9. Conclusion**	Summarize the major findings, key insights, research gap, and future direction.
13	**References**	List all research papers, books, datasets, websites, and other sources using a consistent citation style.

Evaluation Rubrics:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
1. Research Problem Identification and Understanding	10%	9–10% – Clearly identifies the research problem, objectives, scope, and relevance to AI learning approaches.	7–8% – Research problem and objectives are clearly identified with minor omissions.	5–6% – Basic understanding of the research problem with some gaps.	0–4% – Problem is unclear, inappropriate, or poorly understood.
2. Literature Search and Source Selection	10%	9–10% – Uses diverse, highly relevant, recent, and credible research papers, journals, conferences, and technical sources.	7–8% – Uses relevant and credible sources with minor limitations.	5–6% – Uses a limited number of relevant sources.	0–4% – Sources are insufficient, unreliable, or largely irrelevant.
3. Literature Review and Synthesis	15%	13–15% – Provides a comprehensive review and synthesizes findings across multiple studies rather than merely summarizing them.	10–12% – Good review with meaningful comparison of major studies.	7–9% – Basic review with limited synthesis and comparison.	0–6% – Superficial review with little understanding of existing research.

4. Analysis of AI Techniques / Methods	15%	13–15% – Provides technically accurate and in-depth analysis of relevant methods such as decision trees, statistical learning, neural networks, kernel methods, or reinforcement learning.	10–12% – Provides good technical analysis with minor gaps.	7–9% – Provides basic technical description with limited analysis.	0–6% – Methods are poorly understood or inaccurately described.
5. Comparative Analysis of Research Approaches	10%	9–10% – Critically compares approaches based on accuracy, complexity, interpretability, scalability, computational cost, and application suitability.	7–8% – Provides meaningful comparison using several parameters.	5–6% – Basic comparison with limited criteria.	0–4% – Little or no meaningful comparison.
6. Critical Evaluation of Research Findings	10%	9–10% – Critically evaluates strengths, weaknesses, assumptions, experimental methods, and findings of existing studies.	7–8% – Provides good critical evaluation with minor gaps.	5–6% – Some critical observations are provided but analysis is limited.	0–4% – Mostly descriptive with little critical evaluation.

7. Identification of Research Gaps	10%	9–10% – Clearly identifies significant research gaps, limitations, unresolved issues, and opportunities for further investigation.	7–8% – Identifies relevant research gaps with reasonable justification.	5–6% – Identifies basic gaps but provides limited justification.	0–4% – Research gaps are missing or incorrectly identified.
8. Research Insights and Recommendations	5%	5% – Provides insightful conclusions and technically feasible recommendations for future research or applications.	4% – Provides relevant conclusions and recommendations.	2–3% – Basic conclusions with limited recommendations.	0–1% – Conclusions or recommendations are absent or inappropriate.
9. Technical Report and Referencing	10%	9–10% – Report is well structured, technically accurate, professionally written, and uses a consistent citation/reference style.	7–8% – Well-structured report with minor writing or referencing issues.	5–6% – Adequate report but contains several structural or referencing issues.	0–4% – Poorly structured, inadequately referenced, or incomplete report.
10. Originality, Academic Integrity and Use of Sources	5%	5% – Demonstrates original synthesis, proper citation, and strong academic integrity with no inappropriate copying.	4% – Good originality and proper citation with minor issues.	2–3% – Some original analysis but significant dependence on sources.	0–1% – Evidence of substantial copying, poor citation, or academic-integrity concerns.

Decision Tree Learning: A Comparative Review of ID3, C4.5, and CART

Abstract:

Decision tree learning is an important machine learning technique used for classification and prediction. This review focuses on three well-known decision tree algorithms: ID3, C4.5, and CART. The algorithms are compared based on attribute-selection methods, pruning strategies, interpretability, computational complexity, and classification performance. ID3 uses Information Gain, C4.5 uses Gain Ratio, and CART commonly uses the Gini Index for classification. Research shows that decision trees are attractive because their structure is easy to understand and interpret. Experimental studies have also shown differences in tree-building time, tree depth, and classification accuracy among these algorithms. Recent research suggests that improving the greedy attribute-selection process can provide better performance while maintaining scalability.

Keywords: Decision Tree, ID3, C4.5, CART, Classification, Machine Learning

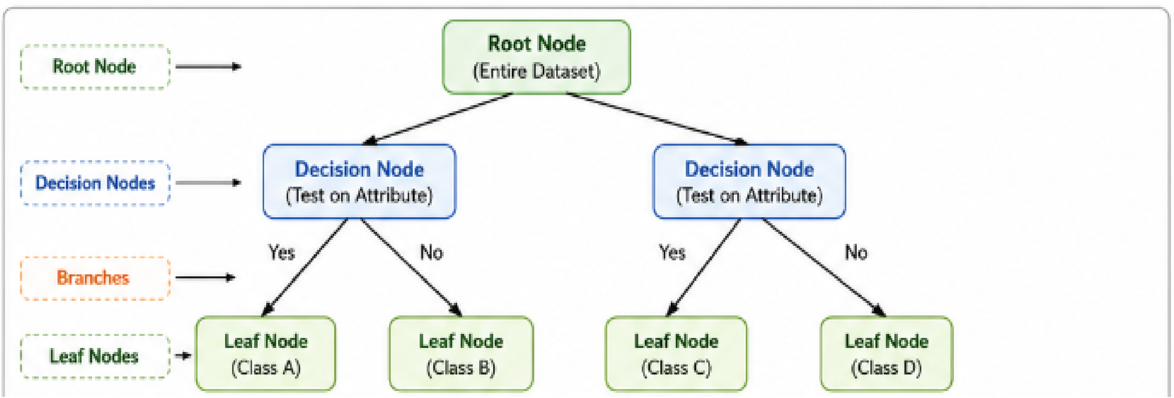
1.Introduction:

Decision tree learning is a supervised machine learning approach used to make predictions by representing decisions in the form of a tree. The internal nodes represent tests on attributes, branches represent outcomes of those tests, and leaf nodes represent the final prediction.

Decision trees are widely used because their hierarchical structure makes the decision process relatively easy for users to understand. ID3 and C4.5 were introduced by J. R. Quinlan, while CART stands for Classification and Regression Trees and is a major decision-tree approach.

The three algorithms use different criteria for selecting the best split. Therefore, comparing them is useful for understanding which approach is appropriate for different datasets and applications.

. Basic Decision Tree Structure



2. Research Problem and Objectives

2.1 Research Problem

Although ID3, C4.5, and CART are widely used decision-tree algorithms, they differ in their attribute-selection criteria, handling of data, pruning methods, computational requirements, and predictive performance. Selecting the most suitable algorithm for a particular dataset therefore requires systematic comparison.

2.2 Objectives:

The objectives of this research review are:

1. To study the working principles of ID3, C4.5, and CART.
2. To compare their attribute-selection methods.
3. To analyze their pruning strategies.
4. To compare their interpretability and computational complexity.
5. To examine their classification performance.
6. To identify limitations and research gaps.
7. To suggest possible improvements for future decision-tree learning.

3. Literature Review

3.1 ID3

ID3, or **Iterative Dichotomiser 3**, is one of the classical decision-tree algorithms. It selects an attribute using **Information Gain**. The attribute that provides the highest information gain is selected for splitting the dataset.

A major advantage of ID3 is its simple construction process and easy-to-understand tree structure. However, the original ID3 approach has limitations when dealing with continuous attributes and can be sensitive to noise.

Research comparing ID3 and C4.5 describes C4.5 as a natural extension of ID3 with improvements to the original approach.

3.2 C4.5

C4.5 is an improved version of ID3 developed by Ross Quinlan. Instead of directly using Information Gain, C4.5 uses **Gain Ratio** for attribute selection.

C4.5 also introduced improvements such as handling continuous attributes and pruning the generated tree. These improvements help reduce some of the limitations associated with ID3.

Research reviews identify C4.5 as an important improvement over ID3 for classification tasks.

3.3 CART

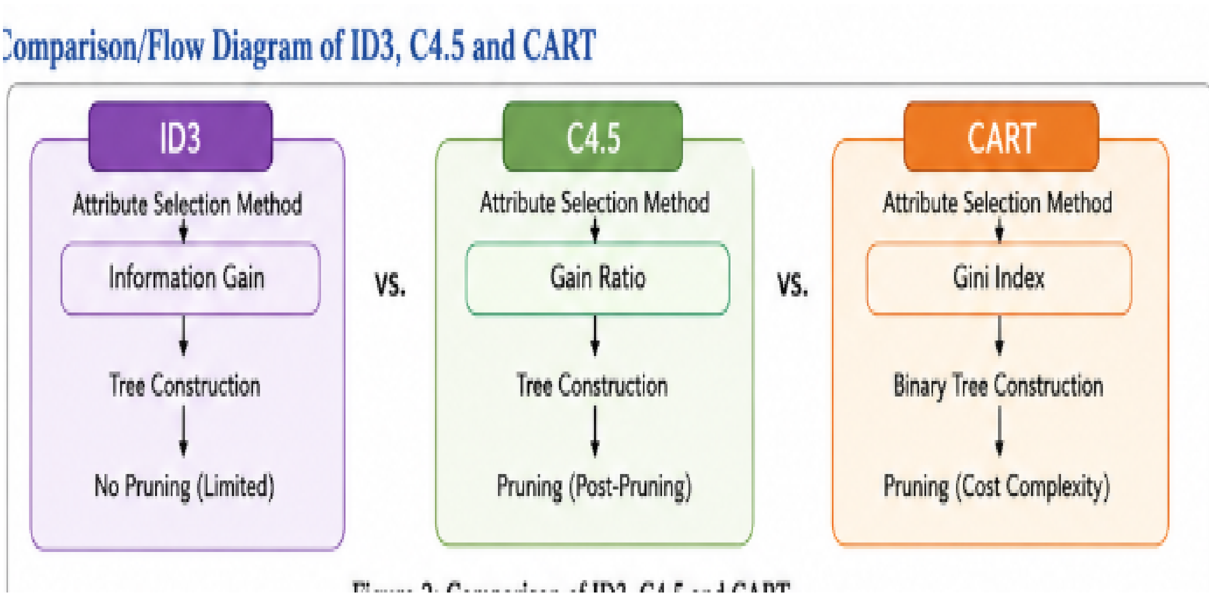
CART stands for **Classification and Regression Trees**. It can be used for both classification and regression problems. For classification, CART commonly uses the **Gini Index** as its splitting criterion.

CART generally produces binary splits, which means each internal decision node normally creates two branches. This can make the resulting tree structured and useful for prediction.

A recent experimental study comparing ID3, C4.5, and CART evaluated them using classification accuracy, tree depth, number of leaf nodes, and tree-construction time.

4. Comparative Analysis

Parameter	**ID3**	**C4.5**	**CART**
Full Form	Iterative Dichotomiser 3	Successor of ID3	Classification and Regression Trees
Main Use	Classification	Classification	Classification and Regression
Attribute Selection	Information Gain	Gain Ratio	Gini Index for classification
Tree Splitting	Can produce multiple branches	Can produce multiple branches	Binary splitting
Continuous Attributes	Limited in original form	Supported	Supported
Pruning	Limited/original ID3 does not emphasize pruning	Post-pruning	Pruning is supported
Interpretability	High	High	High
Complexity	Relatively simple	More complex than ID3	Efficient in many settings
Main Advantage	Simple and easy to understand	Improved handling and pruning	Supports classification and regression
Main Limitation	Can overfit and has limitations with continuous data	More computationally involved	Binary trees can become deeper



Experimental research has found that ID3 and CART can be efficient in tree construction, while performance varies according to the dataset and evaluation measure.

5. Critical Analysis:

Strengths of ID3

- Simple algorithm.
- Easy to understand.
- Uses Information Gain for straightforward attribute selection.
- Suitable for basic classification problems.

Weaknesses of ID3

- The original approach has limitations with continuous attributes.
- It can produce overly complex trees.
- It may be sensitive to noisy data.

Strengths of C4.5

- Improves upon ID3.
- Uses Gain Ratio.
- Supports continuous attributes.
- Includes pruning.
- Suitable for classification tasks.

Weaknesses of C4.5

- More computationally complex than the basic ID3 approach.
- Tree performance can depend strongly on the dataset and parameter choices.

Strengths of CART
- Can handle both classification and regression.
- Uses binary splitting.
- Supports pruning.
- Can produce effective and relatively efficient trees.

Weaknesses of CART
- Binary splitting can sometimes create deeper trees.
- Performance depends on the selected splitting criterion and dataset.

Research using multiple UCI datasets has shown that decision-tree methods can differ considerably in tree depth and construction time, demonstrating that no single algorithm is universally optimal for every dataset.

6. Research Gap

The major research gap is that traditional decision-tree algorithms generally use **greedy splitting**, where the best attribute at the current step is selected without considering enough alternative future splits.

Recent research has investigated alternatives to this greedy strategy. For example, the **Top-k** approach considers several of the best attributes as possible splits instead of considering only the single best attribute. The research reports improvements in accuracy while remaining more scalable than computationally expensive optimal decision-tree approaches.

Therefore, an important research opportunity is to develop decision-tree algorithms that can achieve better predictive performance without greatly increasing computational cost.

7. Proposed Research Direction

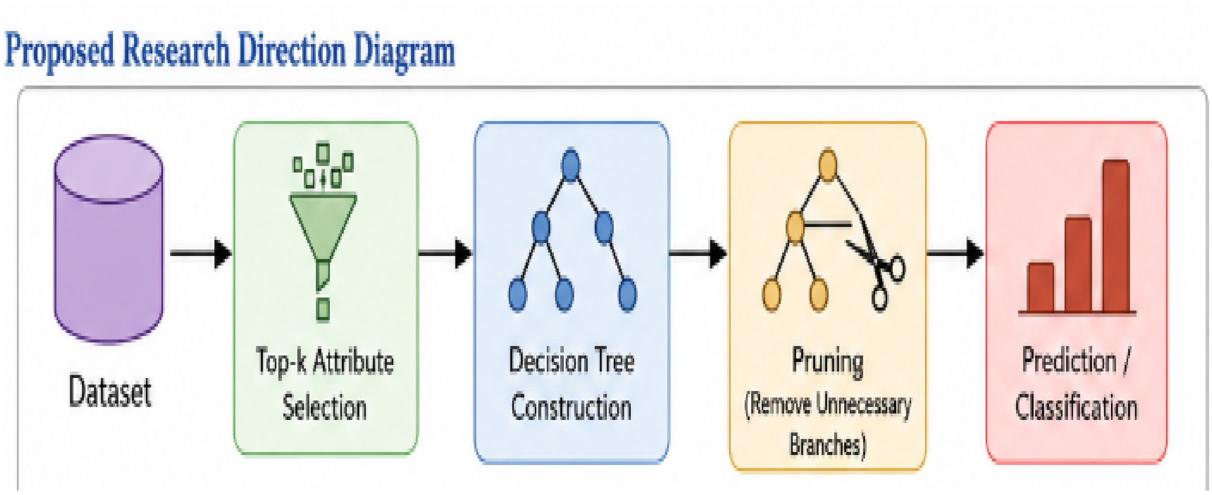
A possible improvement is an **Adaptive Decision Tree using Top-k Attribute Selection and Pruning**.

The proposed approach can:

1. Select the top few candidate attributes instead of only one.
2. Evaluate these candidate splits.
3. Select the most promising split.
4. Continue tree construction recursively.
5. Apply pruning to remove unnecessary branches.
6. Evaluate the final model using accuracy, precision, recall, F1-score, tree depth, and execution time.

This approach could attempt to balance **accuracy, interpretability, and computational efficiency**.

The Top-k decision-tree research provides evidence that considering multiple candidate attributes can improve accuracy while maintaining better scalability than fully optimal tree-learning methods.



8. Discussion

The literature indicates that ID3, C4.5, and CART remain important approaches for understanding decision-tree learning. ID3 provides a simple foundation based on Information Gain. C4.5 extends this idea through Gain Ratio and additional capabilities such as pruning. CART provides a flexible framework for both classification and regression.

Comparative experiments demonstrate that the performance of these algorithms depends on the dataset and the evaluation criteria. For example, studies have evaluated them using accuracy, tree depth, leaf-node count, and construction time rather than relying only on classification accuracy.

Thus, algorithm selection should consider not only accuracy but also computational cost, interpretability, tree complexity, and the characteristics of the dataset.

9.Conclusion

Decision tree learning provides an interpretable and practical approach to machine learning. ID3, C4.5, and CART use different splitting strategies and have different strengths and limitations. ID3 is simple and easy to understand, C4.5 improves upon ID3 through Gain Ratio and pruning, and CART provides a flexible framework for classification and regression.

The review shows that **there is no single decision-tree algorithm that is best for every dataset**. Future research should focus on improving the greedy splitting process while maintaining computational efficiency and interpretability. Adaptive methods such as Top-k attribute selection provide a promising direction for achieving this balance.

References

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